

Interactive Medical Image Segmentation: A Benchmark Dataset and Baseline

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Abstract

Interactive Medical Image Segmentation (IMIS) has long been constrained by the limited availability of large-scale, diverse, and densely annotated datasets, which hinders model generalization and consistent evaluation across different models. In this paper, we introduce the IMed-361M benchmark dataset, a significant advancement in general IMIS research. First, we collect and standardize over 6.4 million medical images and their corresponding ground truth masks from multiple data sources. Then, leveraging the strong object recognition capabilities of a vision foundational model, we automatically generated dense interactive masks for each image and ensured their quality through rigorous quality control and granularity management. Unlike previous datasets, which are limited by specific modalities or sparse annotations, IMed-361M spans 14 modalities and 204 segmentation targets, totaling 361 million masks—an average of 56 masks per image. Finally, we developed an IMIS baseline network on this dataset that supports high-quality mask generation through interactive inputs, including clicks, bounding boxes, text prompts, and their combinations. We evaluate its performance on medical image segmentation tasks from multiple perspectives, demonstrating superior accuracy and scalability compared to existing interactive segmentation models. To facilitate research on foundational models in medical computer vision, we release the IMed-361M and model at <https://github.com/uni-medical/IMIS-Bench>.

1. Introduction

Interactive Medical Image Segmentation (IMIS) enables clinicians or users to guide the model by marking points, lines, or regions on an image, resulting in segmentation outcomes that better meet clinical needs [1, 2]. This user-informed segmentation approach not only optimizes results throughout the segmentation process, aiding clinicians in precisely localizing targets during diagnosis and treatment [3–5], but also addresses the limitations of fully automated segmentation models in generalizing to unseen object classes (i.e., zero-shot learning) [6–11], providing clinicians with a flexible adjustment mechanism that significantly enhances segmentation reliability. However, the advancement of IMIS advancement faces a critical bottleneck: the lack of high-quality, diverse, and large-scale datasets that capture the complex clinical environments [12, 13]. Existing public datasets (such as TotalSegmentator [14], AutoPET [15], and CoNIC [16]) are often tailored to specific modalities or tasks, thus constraining the generalization capabilities of models across diverse medical scenarios.

Recently, as a foundational model for general visual segmentation, the Segment Anything Model (SAM) has garnered widespread attention [17, 18]. SAM integrates simple user interactions (such as points or bounding boxes) into the model’s learning process and leverages pre-training on large-scale datasets to achieve cross-domain and multi-task transferability [19–21]. Although SAM has been pre-trained on over 1 billion natural image masks, its direct application in medical imaging remains limited due to the substantial domain differences between natural and medical images, as well as the lack of medical imaging-specific knowledge. To address these issues, current methods such as MedSAM [22] and SAM-Med2D [23] attempt to expand dataset size and fine-tune SAM to create IMIS models tailored for clinical scenarios. While these approaches have

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shown some progress, their ability to ‘segment anything’ in medical imaging remains limited. This limitation is mainly due to the lack of densely masks in existing medical datasets compared to the SA-1B dataset [17] used to train SAM [22–24]. For instance, the COSMOS dataset [24] contains an average of only 5.7 masks per image, which restricting the model’s capacity for dense segmentation and hindering comprehensive, fine-grained interaction. Moreover, many methods are evaluated only on specific modalities or with limited interaction strategies, further restricting the comprehensiveness and reliability of IMIS model evaluation [22, 24–26].

Addressing these limitations requires the development of a high-quality IMIS benchmark dataset, which is essential for advancing foundational models in medical imaging [4, 27–30]. An ideal IMIS benchmark dataset should meet three core criteria: **(1) Large-scale.** The dataset should be large enough to fully support deep learning model training, enabling the model to effectively capture medical features; **(2) Diversity.** The dataset should encompass various medical imaging modalities and complex clinical scenarios to ensure the model’s ability to generalize across modalities and tasks; **(3) High-quality and densely masks.** Accurate segmentation of complex medical images relies on high-quality annotations, and densely masks further enhance the model’s capability to “segment anything” in medical imaging. Unfortunately, existing public medical segmentation datasets [22–24, 31] do not fully meet these standards, limiting their ability to comprehensively support and evaluate IMIS models.

In this work, we introduce IMed-361M, a benchmark dataset specifically designed for IMIS tasks, this dataset integrates public and private data sources and utilizes foundational models [17] for automated annotation to generating dense masks for each image. A standardized data processing workflow is ensure the high quality and consistency of all masks. IMed-361M achieves unprecedented scale, diversity, and mask quality, comprising 6.4 million images spanning 14 imaging modalities and 204 targets, with a total of 361 million masks, averaging 56 masks per image. IMed-361M effectively solves the problems of small data size and sparse annotations, providing data support for IMIS model training. Additionally, we develop an IMIS baseline model and conduct a comprehensive evaluation of its performance across various medical scenarios, including its effectiveness on different modalities, anatomical structures, and organs, as well as the impact of various interaction strategies on model outcomes. This analysis provided an in-depth understanding of the strengths and limitations of different interactive segmentation methods, establishing a fair and consistent framework for evaluating IMIS model performance. We anticipate that the IMed-361M dataset and baseline model will drive the widespread adoption of IMIS

technology in clinical practice, accelerating the healthcare industry’s transition toward intelligence and automation.

2. Related Work

This section provides an overview of datasets used for various medical image segmentation tasks and reviews the advancements in IMIS algorithms.

Datasets. Due to variations in imaging protocols, medical images often have multidimensional characteristics (e.g., 2D, 3D), making pixel-level segmentation labeling for specific organs or lesions both expertise-dependent and time-consuming. Early work focused on single-organ or single-lesion annotated datasets [14, 15, 15, 32–36]. These single-task datasets provide foundational data for training automatic segmentation models, significantly enhancing model performance in segmenting specific organs or lesions. With the growth of multi-task requirements, multi-organ and multi-tissue annotated datasets [14, 37–39] have emerged, facilitating the exploration of deep learning models in multi-region segmentation tasks [40, 41]. In recent years, visual foundational models have demonstrated impressive cross-domain and multi-task transfer capabilities, motivating researchers to build large-scale, multimodal datasets for medical foundational models. The most common approach is to integrate data from various sources [22–24, 31] to bridge the scale gap compared to natural image datasets. However, due to the sparsity of annotations in the source data, the combined dataset still lags behind natural image datasets in terms of mask quantity and density. For example, the SA-1B [17] dataset averages 100 masks per image, whereas large-scale medical datasets like COSMOS [24] and SA-Med2D-20M [42] have fewer than 5 masks per image on average. This gap limits the ability of medical datasets to support dense segmentation tasks and fine-grained interactions, as well as the applicability of foundational models to “segment anything” in the medical imaging domain.

Algorithms. IMIS methods are generally more effective than automated methods at generating high-quality results that meet clinical requirements, especially in scenarios involving diverse imaging protocols, complex pathological variations, and ambiguous lesion boundaries [43]. User interactions can be take several typical forms, such as scribbles [44, 45], bounding boxes [17, 46], clicks [2, 47], or language prompts [17, 48, 49]. Traditional methods approach this task through energy by minimizing energy on a regular pixel grid, capturing low-level appearance features with unary potentials and promoting consistent segmentation outputs with pairwise or higher-order potentials [44, 50]. With advancements in deep learning, researchers have explored using user prompts directly as network input features, often in combination with 2D or 3D neural networks to produce segmentation results [1, 2, 17, 44–47]. Lever-

aging the advantage of pretraining on large-scale datasets, the Segment Anything Model (SAM) has become a benchmark for IMIS. For example, SAM can serve as a powerful pretrained encoder-decoder model fine-tuned for specific tasks [51, 52]. However, these approaches essentially revert to the design of fully automated segmentation models. Another category of methods retains the interactive aspect of segmentation, focusing on parameter-efficient fine-tuning (PEFT) techniques [53, 54]. However, these methods consider only a limited of interaction strategies, and inconsistencies in evaluation have affected the comparability and reliability of their results. To achieve robust segmentation capabilities across diverse medical imaging datasets, some approaches fine-tune SAM’s decoder using large-scale medical datasets [22–24]. Although these methods demonstrate significant advantages across different medical modalities and tasks, they have not yet been consistently benchmarked against similar methods and lack a thorough examination of how different interaction strategies affect outcomes.

In this work, we evaluate these state-of-the-art methods on the IMed-361M dataset, providing a comprehensive and fair comparison, while also exploring the impact of different interaction strategies on medical image segmentation.

3. IMIS Benchmark Dataset: IMed-361M

We present IMed-361M, the first large-scale IMIS dataset, that significantly surpasses existing datasets in terms of scale, diversity, quality, and density. This section details the data collection and pre-processing procedures, along with an in-depth analysis of the dataset’s advantages and potential applications.

3.1. Data Collection and Pre-processing

Data collection. We integrate over 110 publicly available medical image segmentation datasets from globally recognized platforms such as TCIA¹, OpenNeuro², NITRC³, Grand Challenge⁴, Synapse⁵, CodaLab⁶, and GitHub⁷, covering both 2D and 3D images and variety of formats (e.g., .jpg, .npy, .nii). Additionally, we collaborate with several medical institutions, acquiring diverse clinical data through rigorous ethical review processes, which further enriches the dataset. A detailed list of data sources is provided in the supplementary material.

Preprocessing and filtering. We first standardize all collected medical images according to the SA-Med2D-20M [42] protocol, converting the corresponding ground

truth (GT) into one-hot encoding and storing it in .npz files in compressed sparse row (CSR) format [55]. Then, we apply the following exclusion criteria: (1) exclude 3D slice images and their corresponding masks with an aspect ratio greater than 1.5 to avoid interference from images with excessive geometric distortion during model training; (2) exclude masks where the foreground area accounts for less than one-thousandth of the total pixels, as such small foreground regions are prone to being lost during resizing, making it challenging to generate effective prompts.

Resolving conflicts and ambiguities. To address conflicts and ambiguities in the GT, we first standardize expressions for the same target. For instance, both “lung nodule” and “pulmonary nodule” are renamed to “lung nodule” for consistency. We then manually review and correct misalignments and informational errors in the dataset to ensure annotation accuracy. Finally, for annotations with multiple connected components, we differentiate and label them based on clinical needs to avoid potential misunderstandings that could arise from single-point interactions.

Through this process, we collected 6.4 million images and 87.6 million GT masks that were manually annotated and corrected. Although this already exceeds the data volume of most existing segmentation datasets, it still does not meet our fundamental requirement for the IMIS dataset: diverse and densely masks. Therefore, we further introduce interactive masks to address the above issues.

3.2. Interactive Masks

Automatic mask generation. The automatic mask generation method employed by SAM has been proven to be high-quality and effective [17, 56, 57]. We leverage SAM’s object-awareness capability to generate as many masks as possible for each image. Specifically, we employ a 32×32 point grid to guide the model, generating a set of candidate masks for each point corresponding to potential objects of interest. The generated masks are refined and optimized using the following strategies: (1) Confidence filtering: Only masks with an Intersection over Union (IoU) prediction confidence score above 0.85 are retained to ensure segmentation accuracy. (2) Non-Maximum Suppression (NMS) [58]: For overlapping masks with an IoU greater than 0.7, only the mask with the highest confidence score is kept to eliminate redundancy. (3) Remove background masks: Masks covering more than 80% of the area are discarded, as such large-coverage foreground masks are almost nonexistent in the GT.

Quality control and granularity management. Based on our observations, the generated masks often fail to fully separate structures with unclear boundaries, such as the left atrium, myocardium, and right atrium in the heart. Additionally, masks for dispersed structures like the intestines are often identified as multiple separate objects. To address

¹<https://www.cancerimagingarchive.net>

²<https://openneuro.org>

³<https://www.nitrc.org>

⁴<https://grand-challenge.org>

⁵<https://www.synapse.org>

⁶<https://codalab.org>

⁷<https://github.com>

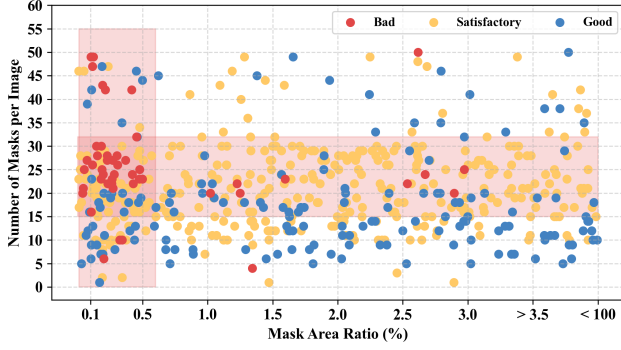


Figure 2. Evaluation of the quality of interactive masks.

subset of IMed-361M (a total of 550 images) and invited four radiologists to evaluate the quality of the corresponding interactive masks. We categorize the mask quality into three levels: (1) “Good”: the mask closely matches the manual annotations; (2) “Satisfactory”: the mask has some visible errors (e.g., incomplete contours) but is generally acceptable; (3) “Bad”: the mask has significant errors (e.g., over-segmentation or under-segmentation) and is unsuitable for clinical segmentation tasks. The evaluation results show that 156 images are rated as “Good”, 345 as “Satisfactory”, and 49 as “Bad”. Statistical analysis reveals that the masks rated as “Bad” mainly come from 18 datasets, concentrated within the red-boxed area in Fig.2. Radiologists note that some excessively annotated structures hold little clinical relevance. Additionally, some masks include elements unrelated to human organs or lesions, such as letters in X-ray backgrounds or hair in dermatoscopic images. Based on this analysis, we manually remove the excessively annotated masks from these 18 datasets and apply a 0.5% foreground filter rate to retain only the final valid masks. Finally, we retain those masks unrelated to human organs or lesions to enhance the model’s adaptability in different scenarios.

4. IMIS Baseline Network

4.1. Model Design

We adopt a strategy similar to fine-tuning SAM [22–24] to establish the IMIS baseline, providing a performance benchmark for future work. As shown in Fig.3, IMIS-Net has three key components: an image encoder for extracting image features, a prompt encoder for integrating user interaction information, and a mask decoder for generating segmentation results using image and prompt embeddings.

We select ViT-base [60] as the image encoder. While larger ViT models (e.g., ViT-Large, ViT-Huge) [22, 24] offer slight accuracy improvements, they significantly increase computational costs, making them impractical for clinical applications. Additionally, most open-source fine-

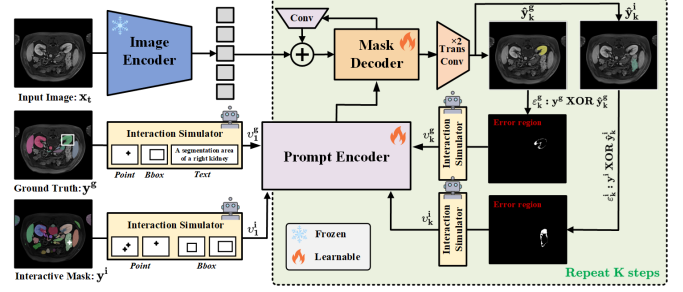


Figure 3. The training process of IMIS-Net simulates K consecutive steps of interactive segmentation.

tuning methods also use ViT-base, ensuring fairness in our baseline. The image input size is $1024 \times 1024 \times 3$, with a patch size of $16 \times 16 \times 3$. For the prompt encoder, we consider three prompt types: points, boxes, and text. Points and boxes are represented by the sum of positional encoding and learned embeddings, while text is encoded using CLIP’s text encoder [61]. Text prompts follow the template: “A segmentation area of a [category]”, covering over 200 organ and lesion categories to establish a benchmark for future multimodal segmentation research. The mask decoder uses Transformer decoder blocks, mapping image and prompt embeddings to the mask. The mask resolution is increased to 256×256 through transposed convolutions and matched to the input size via bilinear interpolation. Notably, expanding the decoder can improve performance without significantly increasing network training parameters, thereby providing a simple solution to address the issue of model performance saturation.

4.2. Experiment Configuration

Training strategy. An overview is provided of the simulated continuous interactive segmentation training [22, 23, 62] in Fig.3. For a given segmentation task and medical image x_t , we first simulate a set of initial interactions u_1^g and u_1^i based on the corresponding ground truth y^g and interactive mask y^i , which include clicks, bboxes, and text input. The click points are uniformly sampled from the foreground regions of y^g or y^i , while the bboxes are defined as the smallest bounding box around the target, with an offset of 5 pixels added to each coordinate to simulate slight user bias during the interaction process. The entire training process involves K interactive training iterations (with $K = 8$ in this paper). The model’s initial predictions are $\hat{y}_1^g$ and $\hat{y}_1^i$. After the first prediction, we simulate subsequent corrections based on the previous predictions $\hat{y}_k^g$ and $\hat{y}_k^i$, as well as the error region ε_k between the y^g and y^i , where $k \in \{1, \dots, K\}$. Additionally, we provide the low-resolution predicted mask from the previous prediction as an extra cue to the model. As can be seen, the image encoder only needs to encode the image once during the training, and subse-

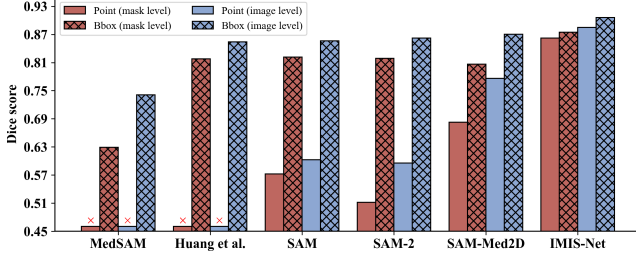


Figure 4. Comparison of IMIS-Net with existing foundation models, with performance statistics at both image and mask levels.

quent interactive training only updates the prompt encoder and mask decoder parameters.

Training setting. We use the Adam optimizer [63] with a learning rate of 2×10^{-5} . The training was conducted on 72 NVIDIA 4090 GPUs with a batch size of 2. For each image, we randomly select 5 targets from the corresponding GT and interactive masks as supervision targets (if fewer than 5 targets are present, they are selected repeatedly). The image resolution was uniformly resized to 1024×1024 , and pixel intensities are randomly scaled and shifted with a probability of 20%, where the scaling factor and offset are set to 0.2. We train the model for a total of 12 epochs on the IMed-361M dataset and select the final checkpoint as the model weights. A linear combination of Focal loss [64] and Dice loss [65] is used as the loss function, balancing their influence in a 20:1 ratio. Finally, the Dice score is used as the primary evaluation metric for this study.

4.3. Evaluation Model and Data

Model. We evaluate five visual foundation models [17, 18, 22–24], selected based on their pretraining on large-scale datasets and the availability of source code. Among them, MedSAM, SAM-Med2D, and Huang et al. [24] are specifically designed for IMIS, while SAM [17] and SAM-2 [18] are pre-trained on massive natural image and video datasets. To ensure fairness in the evaluation, all models used ViT-Base as the encoder. It is important to note that, MedSAM and Huang et al. [24] only support bbox-based interactivity (using the models released by the authors), while the other models support points, bboxes, and combinations of both.

Data. The data used to evaluate the model is divided into two categories: the test set and the external dataset. The test set contains 58,411 images, with a distribution consistent with that of IMed-361M. The external dataset includes data from three different modalities and clinical scenarios: SegThor [66], which is used for segmentation of thoracic organs at risk in 516 CT images; TotalSegmentator MRI [37], which is used for segmenting organs in 2,147 MRI images; and ISLES⁸, which belongs to the MRI FLAIR modality

⁸<https://www.isles-challenge.org/>

and includes 364 images of ischemic stroke lesions. These datasets were not used during the training of IMIS-Net.

5. Results

5.1. Main Results

Fig.4 shows the performance evaluation of IMIS-Net compared to other vision foundation models on the single-interaction segmentation task. During testing, all models used the same points and bounding boxes as prompts. The results indicate that IMIS-Net outperforms other models in both image and mask-level statistics. The bbox interaction of different models always outperforms the click interaction because the bbox can provide more boundary information. Notably, despite being pretrained on large-scale medical image datasets, MedSAM and SAM-Med2D still exhibit significant performance differences. This disparity is closely related to the scale and diversity of the pretraining datasets. As shown in Fig.5(a) and (b), the pre-training dataset for SAM-Med2D lacks samples of skeletal structures, resulting in poorer segmentation performance for these anatomical structures. Additionally, under the single-point prompt condition, SAM and SAM-2 achieve only Dice scores of 60.26% and 59.57%, respectively. likely due to the absence of medical knowledge in the pretraining data and limited interactive information constraining model performance.

We increase the number of interactions from 1 to 9 to observe the performance changes of the model (Fig.5 (c)). As expected, performance improves with more interactions, and the gap between models narrows. This is because multiple interactions reduce the task difficulty. We also examine the impact of click position and bbox offset on performance. As shown in Fig.5 (d) and (f), when the prompt point is closer to the centroid, the performance improvement is more significant, with SAM-2’s Dice score increasing by 2.84%. Additionally, with bbox offset, all methods experience a performance decline of 0.85%-3.94%. Our model exhibited the smallest performance drop, making it more robust for practical use. In summary, leveraging the rich data and mask diversity of the IMed-361M dataset, our model delivers the best performance across various medical scenarios and interaction strategies. More experimental results and analysis can be found in the supplementary material.

5.2. External Dataset Evaluation

Tab.1 presents evaluation results on external datasets. IMIS-Net achieves the best average performance across datasets from three different sources and tasks. We highlight the Dice scores of the 12 major abdominal organs in the TotalSegmentator MRI dataset, where IMIS-Net outperforms in 10. On the ISLES dataset, focusing on ischemic stroke

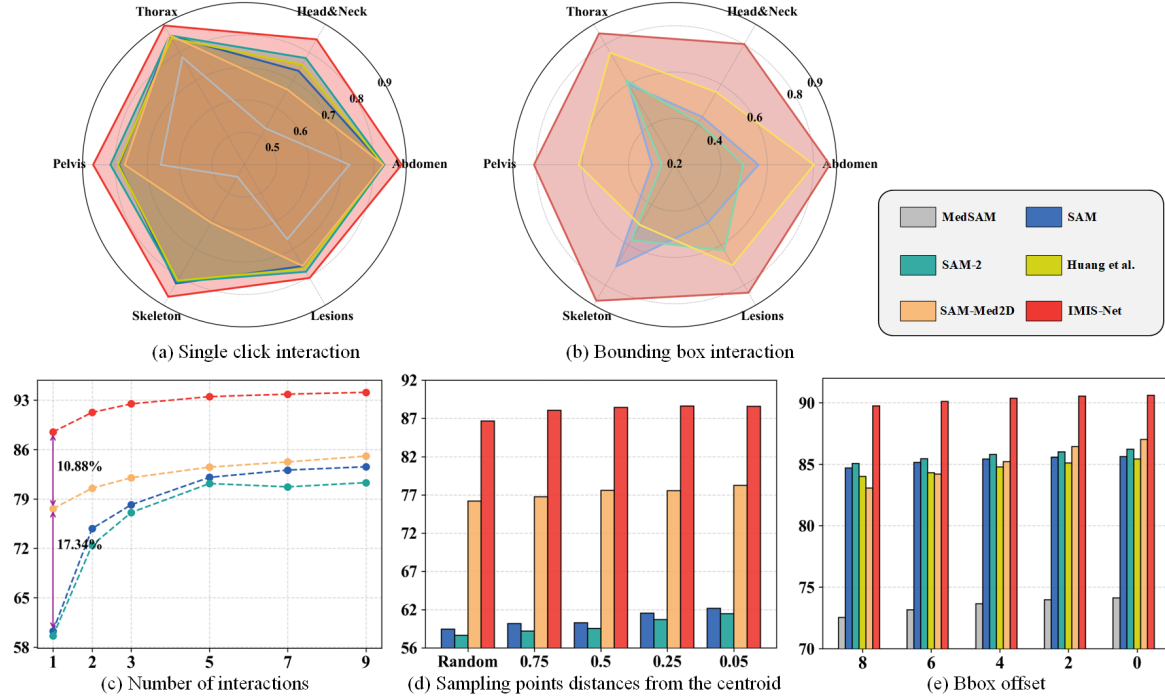


Figure 5. Comparison of segmentation performance across different anatomical structures under single-click (a) and bounding box (b) interactions. (c) Changes in segmentation performance with increasing interaction numbers. (d) and (e) Impact of click position and bounding box offset on performance.

Dataset	Category	SAM	SAM 2	MedSAM	Huang et al.	SAM-Med2D	IMIS-Net
ISLES	Ischemic Stroke Lesion	55.92	60.14	59.90	56.79	68.22	71.78
	Esophagus	78.19	86.60	47.17	76.48	82.04	89.17
	Heart	91.41	92.34	81.16	92.02	94.18	81.27
	Aorta	82.36	83.69	55.68	84.57	79.12	94.92
	Trachea	93.76	94.24	72.40	94.60	85.10	90.04
	Average	84.46	85.86	60.55	84.73	86.43	89.27
SegThor	Adrenal gland	77.19	82.25	56.58	68.52	66.35	77.53
	Aorta	76.88	79.90	55.98	80.08	74.91	85.62
	Colon	56.69	56.00	42.53	55.41	51.44	73.47
	Duodenum	69.30	74.81	55.64	72.38	69.67	75.97
	Gallbladder	81.33	84.82	68.15	83.46	87.63	82.93
	Iliopsoas	59.57	71.96	58.48	63.67	60.76	72.55
	Kidney	79.43	81.59	58.41	81.09	74.04	82.72
	Inferior vena cava	84.94	87.03	57.73	85.88	78.34	87.35
	Liver	86.93	88.70	68.14	87.35	86.75	90.62
	Pancreas	66.95	69.86	46.10	66.30	66.84	70.84
	Spleen	81.12	83.05	66.61	83.87	84.95	86.74
Totalsegmentator MRI	Stomach	77.10	81.23	62.75	76.52	79.31	81.45
	Average	75.45	77.62	59.52	75.47	75.92	79.06

Table 1. Quantitative comparison results of IMIS-Net against five other interactive segmentation methods on three external datasets.

lesion segmentation, our model surpasses the second-best SAM-Med2D by 3.56%. These results demonstrate the strong generalization ability of our method.

5.3. Ablation Study

Scaling Up Training Data. As shown in Fig.6 (a) and (b), when trained only on the GT from IMed-361M, IMIS-Net performs poorly. However, with the addition of interac-

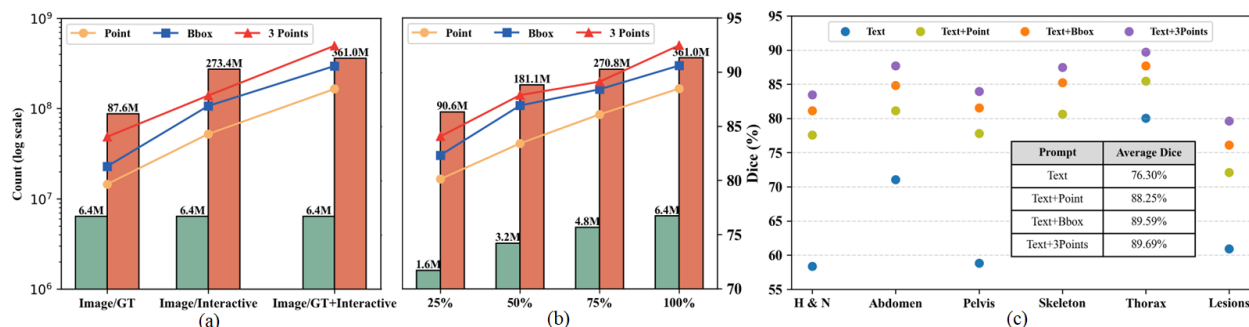


Figure 6. (a) and (b) segmentation performance increases with the increasing mask density or data scales up. (c) effectiveness of text and its combination with other prompts.

Decoder dimension	Image resolution	Prompt		Training parameters
		Point	Bbox	
768	256×256	0.8214	0.8469	29.68 M
768	512×512	0.8673	0.8968	29.68 M
256	1024×1024	0.8366	0.8497	5.52 M
512	1024×1024	0.8563	0.8729	15.19 M
768	1024×1024	0.8848	0.9060	29.68 M

Table 2. Ablation study of model design, including decoder dimension and image resolution.

tive masks, the Dice score increases rapidly. Similar performance trends are observed when training on datasets of different sizes, indicating that our method is scalable and performs better with more available training data.

Text and Combination Prompts. IMIS-Net, as a versatile interactive model, can utilize text prompts to achieve the segmentation of over 200 organs and lesions. As shown in Fig.6 (c), the model achieves a segmentation performance of 76.30% when using only text prompts. When both text and point prompts are combined, the average Dice score increases by 11.95%. Furthermore, after three rounds of click-based correction, the Dice score reaches 89.69%. These results show that combining different prompts synergistically enhances the model’s segmentation capabilities.

Model Design. We conduct an ablation study on different configurations of the model, including the impact of input resolution and decoder dimension on model performance. As shown in Tab.2, the model’s segmentation performance improves as the training image resolution increases. This higher input resolution allows for clearer visualization of lesions and organs in medical images. Additionally, when the decoder dimension is increased from 256 to 768, the model’s performance improves from 84.97% to 90.60%, with only a 24.16M increase in trainable parameters. These

results demonstrate the scalability of IMIS-Net, showing that even when the model’s performance approaches saturation on larger datasets, further performance improvements can still be achieved by expanding the decoder.

6. Conclusion

In this work, we introduce IMed-361M, a benchmark dataset dedicated to interactive medical image segmentation. It includes a vast array of medical images across various modalities, extensive segmentation scenarios, and densely masks, surpassing all existing datasets that are limited to single tasks or simple integrations. Leveraging this data resource, we developed a general IMIS baseline model that enables users to generate segmentation results tailored to clinical needs through interactive methods, including clicks, bounding boxes, text prompts, and their combinations. We performed a comprehensive comparison of this baseline with existing foundational models, demonstrating that our model provides significant performance advantages and exhibits strong transferability in previously unseen scenarios. Notably, our approach typically requires fewer interactions to achieve comparable performance, enhancing its practicality in real-world applications.

The IMed-361M dataset will strongly facilitate the development of foundational models in medical imaging and lay the foundation for fair evaluation across different models. IMIS-Net provides general technical support for various clinical applications, accelerating the widespread application of AI technology in the medical field. Despite these achievements, we recognize that this work still faces several challenges. For instance, effectively obtaining semantic information for interactive masks and extending this approach to more comprehensive and finer-grained medical image analysis scenarios are areas that require further exploration and improvement in the future.

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